Run Tracker Application

A microservice-based application for storing runs data and calculate statistics for them.

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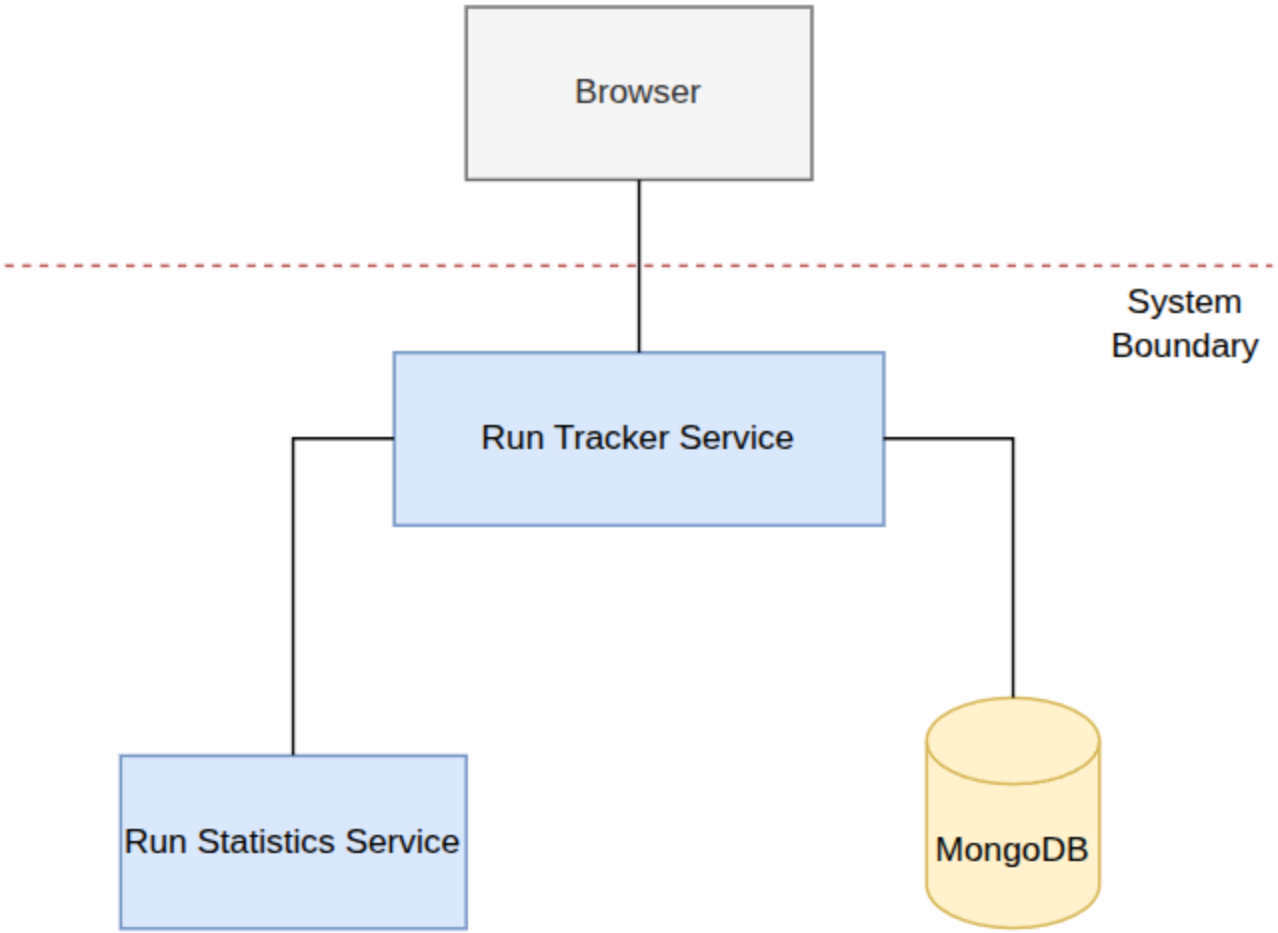
Assignment 1

PA2577 HT26 lp12 Applied Cloud Computing and Big Data

Blekinge Institute of Technology



Architecture – High Level Design



Run Tracker Service (Run Service)

- Main entry point of the application
- Stores and retrieves runs
- Stateless
- Accesses MongoDB directly
- Requests statistics from the Statistics Service
- Serves the web interface

Run Statistics Service

- Calculates statistics for runs
- Stateless

MongoDB

- Stores run data persistently
- database service

Browser / User Interface

- Accesses the application from outside Kubernetes
- Communicates with Run Tracker Service

Architecture – REST API End points

Run Tracker Service

- `GET /runs`

Returns all stored runs

- `POST /runs`

Stores a new run with its data

- `GET /statistics`

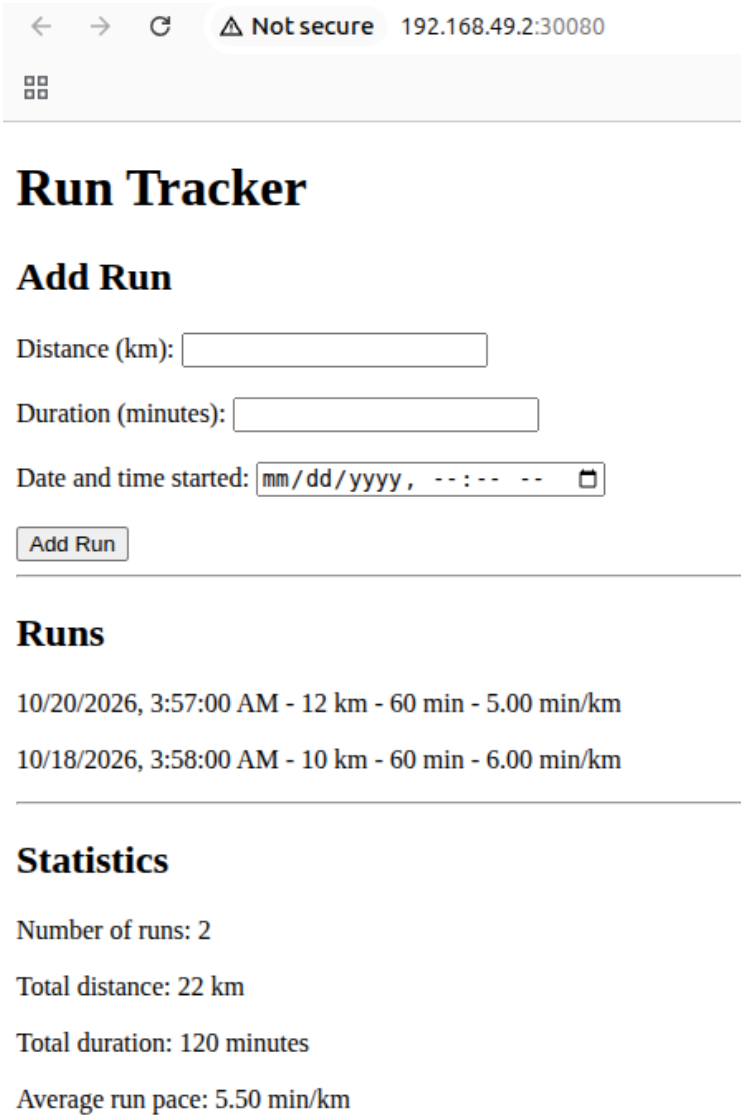
Retrieves runs, sends them to the Statistics Service, and returns the calculated statistics

Run Statistics Service

- `POST /statistics`

  Receives a list of runs and calculates:
  - number of runs
  - total distance
  - total duration
  - average run pace

Demo


Not secure
192.168.49.2:30080
Run Tracker
Add Run
Distance (km):
Duration (minutes):
Date and time started:
mm/dd/yyyy, --:-- --
Add Run
Runs
10/20/2026, 3:57:00 AM - 12 km - 60 min - 5.00 min/km
10/18/2026, 3:58:00 AM - 10 km - 60 min - 6.00 min/km
Statistics
Number of runs: 2
Total distance: 22 km
Total duration: 120 minutes
Average run pace: 5.50 min/km